C7 / SCREENS AND REPORTS

Database functionality evidence

This evidence combines actual compiled Qt screens with independent exported SQL result tables. All data is fictional and uses isolated test storage. The screenshot capture uses Qt widget rendering on the offscreen platform rather than a live desktop capture.

Creation and data entry

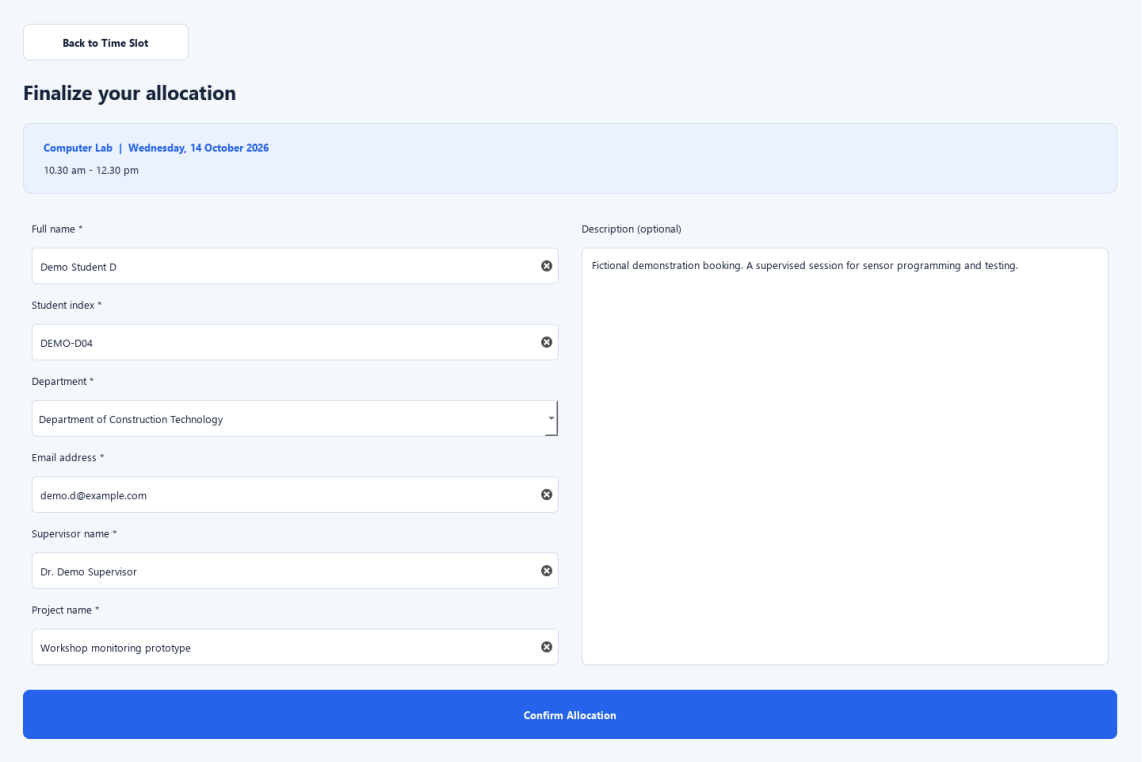


Figure 1. Populated allocation form with selected date/slots and required student, supervisor and project inputs.

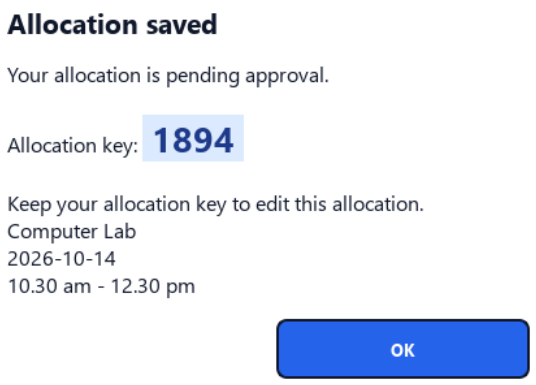


Figure 2. The actual success dialog displays the saved allocation key.

The test harness invokes the active submit path and checks that the database contains the saved record and related slots. See T10 in the runtime test log.

IMPLEMENTED ADMINISTRATOR OPERATIONS

Review, filtering and changes

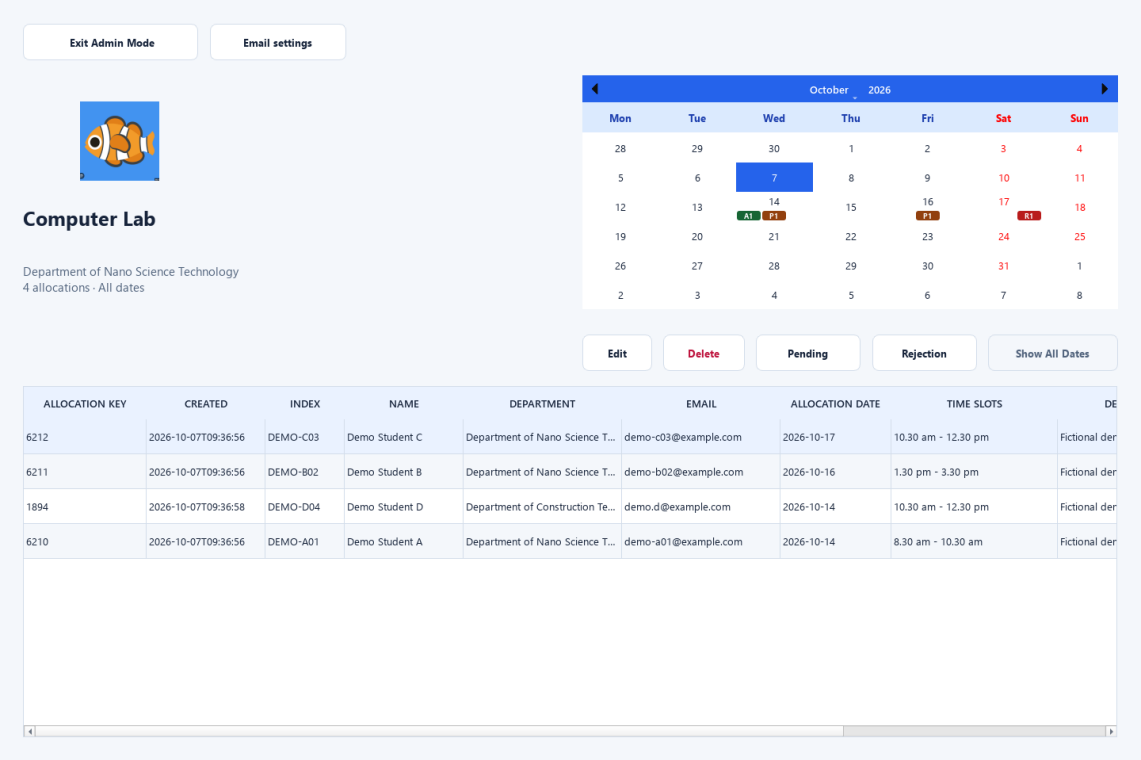


Figure 3. Administrator table showing fictional identity rows and record actions.

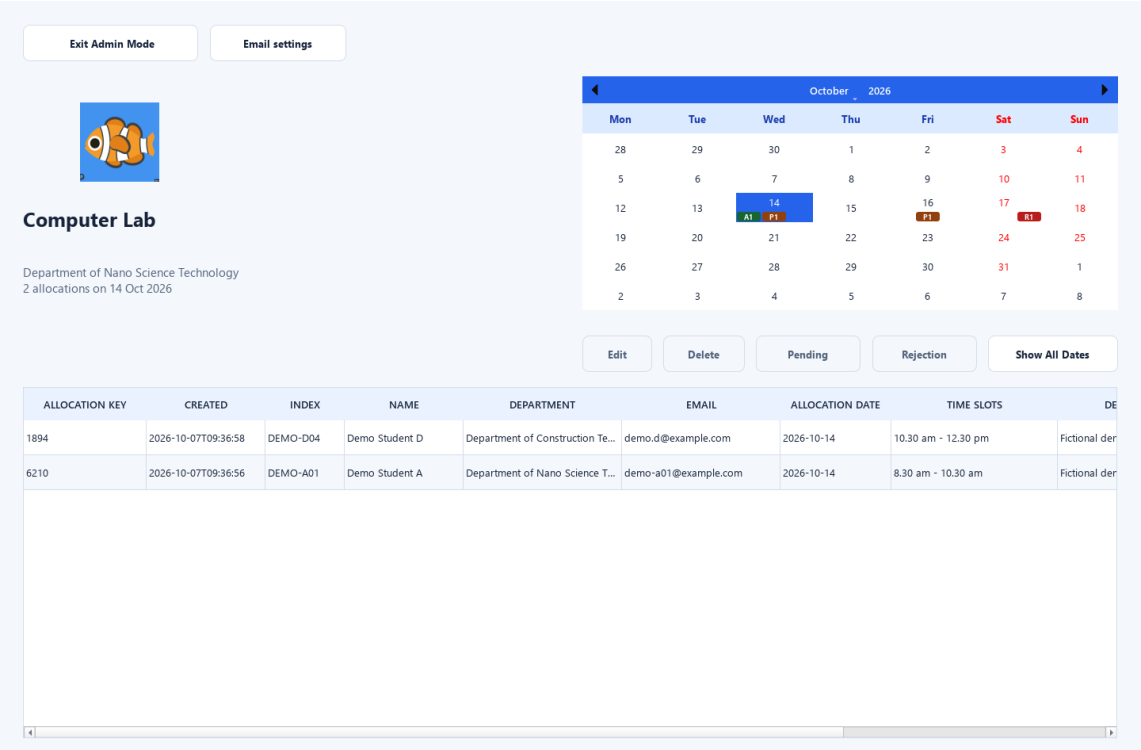


Figure 4. Date filter applies to the current workshop. Full test results also verify record update, deletion and child-slot cascade.

Update and deletion are established by executed database assertions and the saved logs. A screenshot alone does not prove a deleted record or a persisted update.

ADMINISTRATOR REVIEW EVIDENCE

Status controls and persistent updates

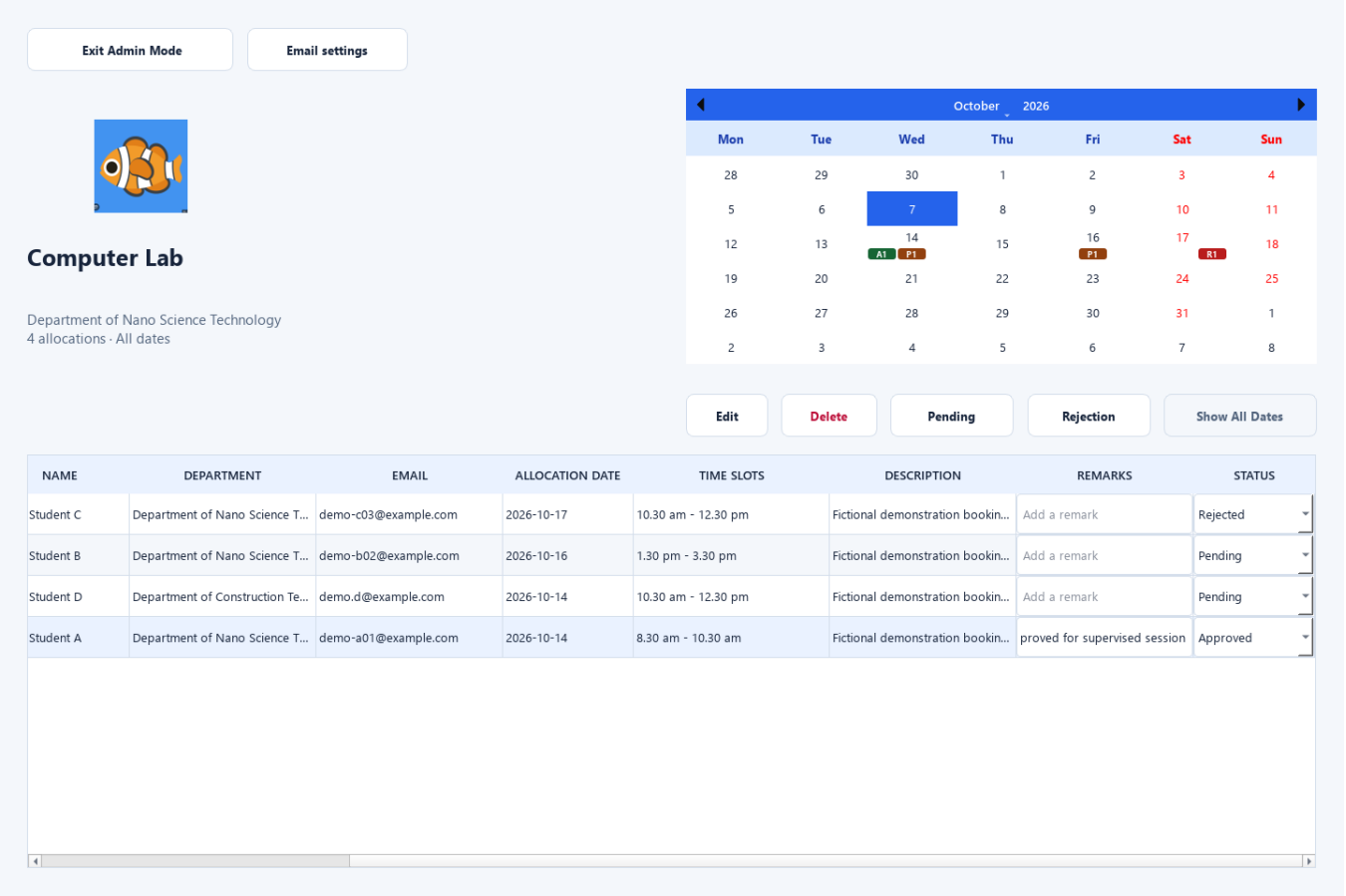


Figure 5. Actual dashboard scrolled right. Remarks and Pending/Approved/Rejected controls are visible. The selected row enables record actions.

Executed update and deletion results

Test	Input / actual observation
T05 Update	Name/project changed and two slots replaced. Original key 4101, created_at and admin remark preserved. Old slot became available.
T06 Status	Rejected record released slots. A second booking reused capacity. Reapproval failed and original Reject status remained.
T08 Delete	Existing record removed. Child slot count became 0. A repeat deletion returned the refresh instruction.

Full expected/observed values are recorded in runtime_results.json. Screenshot values and API assertions use related isolated demo/test datasets rather than real student records.

SUCCESSFUL RECORD UPDATE

Actual UI update: before and after

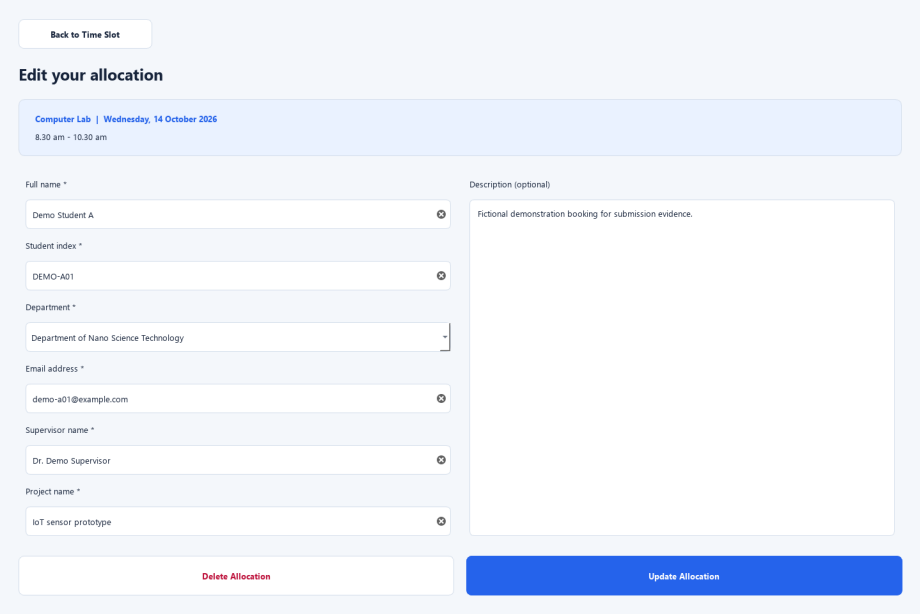


Figure 6. Before update: the actual edit form loads IoT sensor prototype for the approved demonstration record.

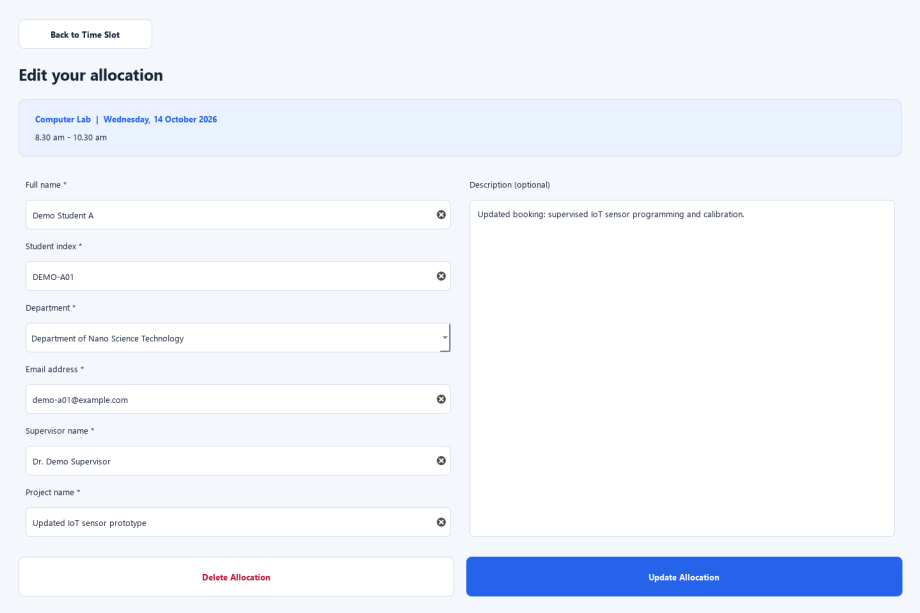


Figure 7. After Update and reopening Edit: Updated IoT sensor prototype and the changed description are read back from SQLite.

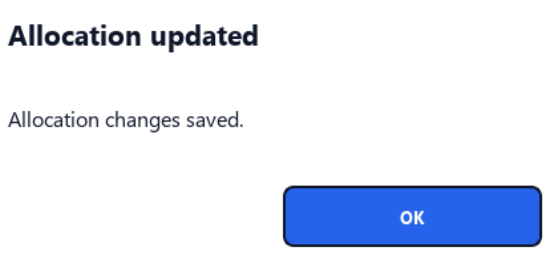


Figure 8. The actual Allocation updated dialog. This additional workflow was executed on crud_demo.sqlite, a copy of the preserved fictional demo database.

SUCCESSFUL RECORD DELETION

Actual UI deletion: before and after

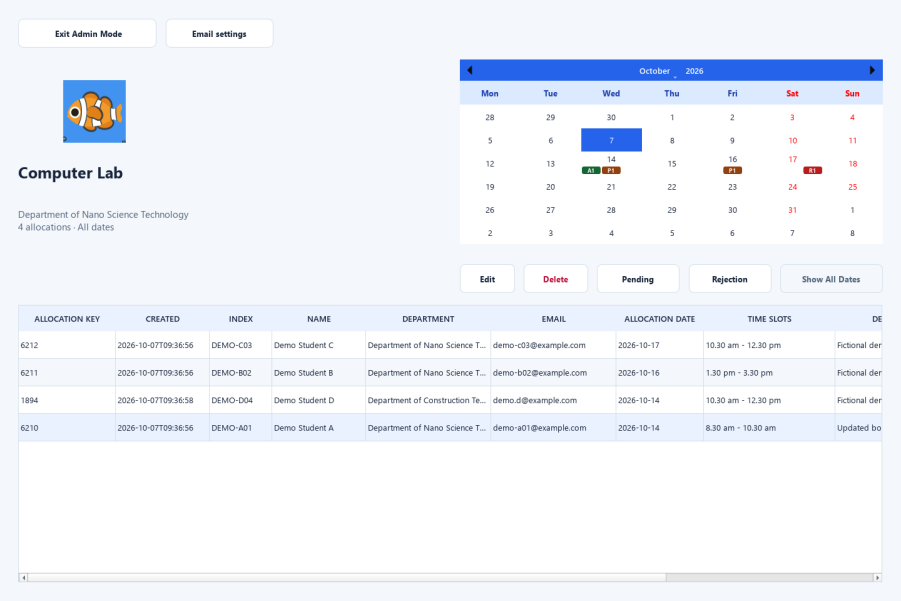


Figure 9. Before deletion: four allocations are visible and key 6210 is selected.

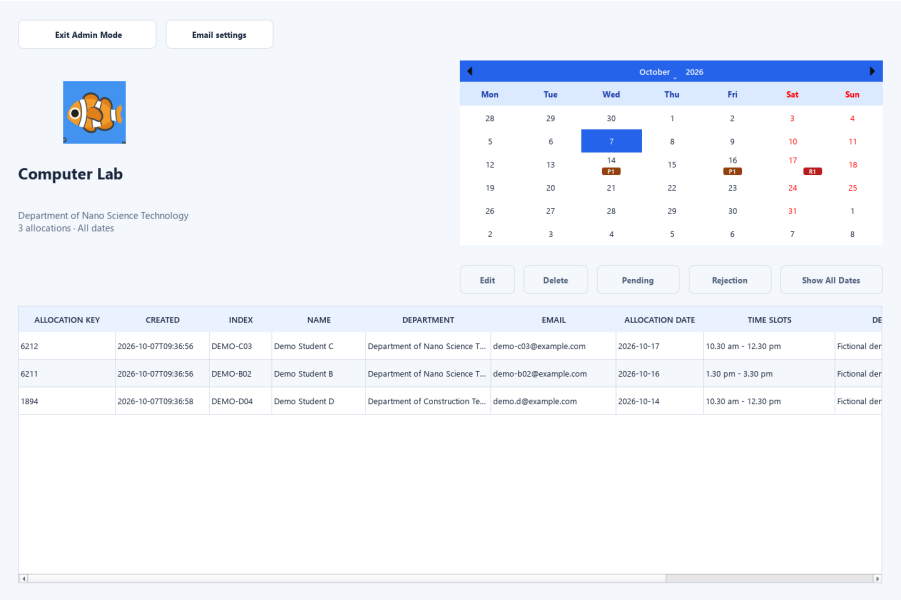


Figure 10. After confirmed Delete: three allocations remain. Key 6210 and its approved calendar badge are absent.

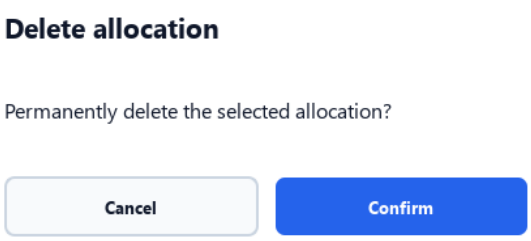


Figure 11. The real deletion confirmation. Additional UI checks confirm zero matching bookings and zero child slot rows. All 19 additional workflow assertions passed; see ui_crud_results.json.

VALIDATION AND ACCESS FEEDBACK

Handled error cases

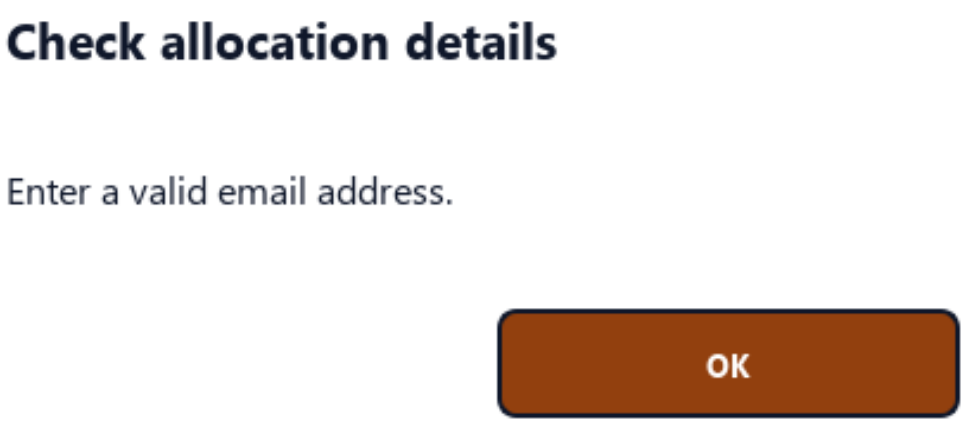


Figure 12. A malformed email opens the application validation dialog. No booking is created.

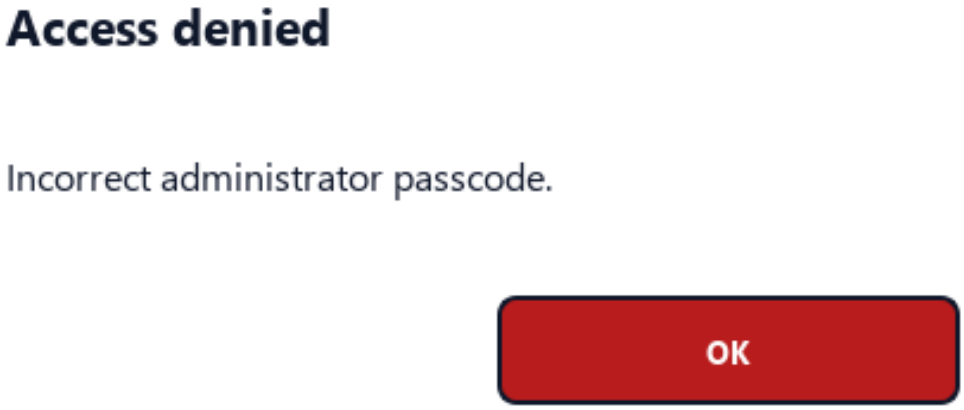


Figure 13. A wrong workshop administrator passcode opens Access denied.

Evidence interpretation

Both images come from executed handlers in the compiled application. The associated tests check that invalid input does not persist and access does not open the administrator view. SQL constraint failures, occupied-slot protection and status conflict rejection are also recorded in logs.

REPRODUCIBLE OUTPUTS

Exported reports and traceability

Evidence item	Observed result
Production build	Qt 6.11.2 / MinGW 13.1.0 release build completed.
Compiled integration test groups	10 / 10 groups passed; 69 assertions; 0 failed assertions.
Independent SQL verification	20 / 20 schema, constraint, integrity and cascade checks passed.
Joined booking output	8 rows
Workshop utilization	10 rows, including unused workshops
Available Computer slots	2 rows for 20 October 2026
Status workload	3 status rows
Student allocation history	3 rows for DEMO/NT/001

How to reproduce

Follow the section C8 README to run the A3 CREATE/INSERT and five SELECT scripts into a fresh output directory. CSV/JSON retain full columns. C7 screenshots also include 17_SYNCLAB_Group_01_SQL_Report_Output.png, an actual executed workload report excerpt; it is external evidence, not an in-app export.

Limits of this evidence

The tests verify the submitted source with isolated data and settings. They do not establish email delivery, restoration on a different computer, every display size, accessibility, long-running load or every legacy database migration. The live demonstration remains a group presentation activity.